Limits to Commute: The Case of Indian Women

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Abstract Research has shown that there is an inherent physiological and psychological difference between men and women resulting in differing commute patterns. Using the household level data from the 76th NSSO round on 'Drinking Water, Sanitation, hygiene and housing conditions;' The present research sheds light on whether the hypothesized commute differences are apparent. Additionally, the research investigates whether the commuting pattern of women is different in urban and rural settings. Further, the research tries to understand the determinants of observed commute patterns in women. This analysis is carried out by subjecting the data to stratified regression. The results are important for developing women centric policies.

Keywords Commute · Rural · Urban · Income · Similarity · Stratified regression

1 Introduction

Women are an important part of the workforce. Transport and gendered division roles are not only correlated, but women's opportunities and accessibility to the available prospects are adversely affected. Gender specific transport attributes such as the resource and time poverty faced by women, their work participation rates in urban and rural areas, high concentration of women in the informal and home-based production sectors etc. are more often than accepted as anecdotal evidence. This intersectional field of gender and mobility suffers from the constraint of availability of data. This constraint is deeply reflected in the research, policy and subsequent implementation of policy. While past investigations confirm that a mix of socio-demographic, economic, transport infrastructure and area characteristics may be helpful in capturing and explaining the commuting pattern, however this has not

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reduced the gender bias in transport policies [1]. Further, this bias is percolating in fast urbanizing developing nations like India. Certain important questions like How far do women commute?, What influences these commuting decisions? etc. remains unanswered.

After setting the need and significance for the research, in the next section, the literature on commute patterns is reviewed. In the third section, the methodology and the data sources are detailed. The fourth section examines whether men and women have identical commute patterns. In the fifth section, we investigate the variables that influence commute patterns. In the sixth section, we summarize our results. Additionally, we conclude by listing a few shortcomings of our article and the path ahead.

2 Literature Review

While past investigations confirm that a mix of socio-demographic, economic, transport infrastructure and area characteristics is helpful for capturing and explaining the commuting patterns in Indian rural–urban megaregion [2], opportunities and accessibility to the available opportunities are adversely affected. It is important to highlight this aspect in order to introduce a transport policy that integrates women's transport needs and which is sensitive to the effects of poor accessibility on the lives of the poor, it is only possible to carry out an investigation into the existing trends, as it is difficult to analyse these trends and suggest a plausible pattern based on the limited data available. The said study tries to investigate transport infrastructure to arrive at a conclusion on commuting patterns in an urban–rural megaregion only to ignore several other underlying parameters that determine the said patterns. For example, the study identifies the relationship between high income women and private vehicle usage among women to be positive, but it is merely correlational and does not define a causal relationship. In the same manner, the positive correlation in respect of high skilled workers and intermediate trips, income and increased private vehicle usage of women has been mistaken for a positive impact, i.e., a determinant. The word 'impact' implies a causality and not a correlation.

While other studies recognize the constraints faced by women with respect to commute, the underlying argument that "commute is determined by so and so" is still conjectural due to anecdotal evidence. Gender is one of the most important variables for mode choice [3]. The differences in travel behaviour by gender are mostly due to the complexity of activities more often experienced by women than men. To what extent does the complexity of activities affect commuting trips is not statistically defined to arrive at conclusions [4].

While it is difficult to factor every constraint faced by women, it becomes important to identify those variables that are statistically significant for commuting trips. The study by Jain [2] uses rail and road density in particular to understand commuting patterns. It proposes an increase in rail infrastructure while there is no objective evidence that it is necessary, when in fact the current rail infrastructure is underutilized.

Anand and Tiwari [4] explore the above constraints objectively, with the same setting of the urban-rural megaregion of Delhi, but due to a lack of information, no conclusion can be deemed fit enough. But one key derivation of the study shows that the lack of mobility of poor women (which is the premise) has a trichotomic effect: it is *caused by their poverty*, *is an indicator of their poverty*, and in *turn causes their poverty.*

Gender inequalities result in the social devaluation of women who are assigned a lower position in society. Gender interacts with transport systems to reinforce inequalities between women and men. Transport infrastructure and services play a facilitative role in women's and men's mobility and access; however, they can also impose physical restrictions on their ability to move [5].

In a country like India, social dynamics to a large extent have suffered due to the amalgamation of caste, creed and religion and its effect on the economic structure. Uteng and Cresswell [5] explain that the removal of women from spaces, due to their social exclusion by virtue of their belonging to a particular caste or creed, causes further social exclusion by distancing them from the physical society. This removal of the physical body from the socio-spatial realm results in an exodus of the physical removal from the inner city. Proper women citizens come into being through their belonging to spaces of normal femininity and respectability and are out of place in inner-city spaces where some other women belong [5]. The notions of race, religion and caste reflect socially in terms of respectability and femininity. These subjective factors cause, indicate and result in constrained mobility in India. However, in the developing stage that the country is in, where there is the constraint of income, and lack of security of employment, the above mentioned factors(caste, religion and creed and hence respectability) are not statistically significant in the context of mobility.

It is also worthy to notice that income and mobility for women is a bidirectional concept. Income restricts or promotes mobility, and mobility restricts or creates income. For women, income and ownership of vehicles are not as statistically significant as they are for men. Factors like safety play a very important role, and such factors differ in degree from one region to another, and so do the expectations and behaviour patterns. So it becomes difficult to pinpoint on a certain set of determinants that apply for all regions, and it becomes crucial to know how much of one determinant restricts the movement of women in their region.

On the periphery, it becomes necessary to look at housing arrangements of both the genders in urban and rural areas, distance from work, family setup, whether the family setup is car or income deficient and several other parameters while drafting a transport policy that is not only women-friendly but also sustainable for the environment [6]. Manoj et al. [7] states that housing tenures of working women in cities like Bangalore show precedence of owned establishments in commuting longer distances. Majority of transport policy is ex-post planning. As people's income rises, they prefer greater floorspace. There is always a growing market for high-end buildings on the periphery of large cities. As is the norm, wherever there is a unit of well-off people residing, a cluster of low income people move in search of low income employment. But the mode of transport and the constraints faced will not be the same for the two. Hence we have different travel behaviours exhibited by people living in the same region/locality.

There seems to be a dearth of structured investigations in understanding the statistical significance of various factors that affect commute, especially in the conjunction of women and commuting patterns. Some studies use the size of a particular area(urban or rural) as a gradient on the scale of small, mid and large to define how the commuting patterns remain the same in comparison to other developing countries. Advani and Mahadevia, [8] state that the proportion of 'no choice trips' or 'forced trips' is much higher among women due to the double disadvantage of constraints they face, and poorly planned transportation schedule/

transport infrastructure. However, it should be kept in mind that with geographic and demographic diversity in India, other mid-size towns in the same country may show patterns distinguishable from each other. Rogers and Srinivasan [9] show that there is a significant difference in travel behaviour because two localities in the same city show marked differences in accessibility.

Therefore, different determinants of travel behaviour for women exhibit different statistical significance in different cities, and sometimes even in different localities in the same city. Because the distance travelled is a function of factors, both measurable(and measurable objectively) and immeasurable(subjective). Women on average also travel less often and for shorter distances than men [10] and are more willing to reduce vehicle use than men [11]. The differences in travel behaviour by gender are mostly due to the complexity of activities more often experienced by women than men. However, the said studies on understanding the constraints experienced by women use anecdotal evidence, which reduces the conclusion to being merely conjectural.

Although there are discernable patterns in the determinants of transport choice for commuting trips across different continents among the two genders still varies due to the constraints that the commuters face, which are experienced differently in different countries. It is now widely acknowledged that transport policies are gender biased in as far as they fail to accommodate the specific gender-based needs of the transport users [1].

3 Objectives

From the literature, certain important questions like whether men and women have similar commute patterns, whether their patterns vary across rural and urban landscapes, or are their other determinants which influence the commute pattern of women remained unanswered. The present research is a modest attempt to understand this. For addressing this overarching question, we maintain the following objectives:

> Ascertaining whether commuting patterns are identical for men and women
> Investigating the influence of income on commute distance
> Examining the influence of socio-economic characteristics on commute distance
> Identifying the influence of family size on commute distance
> Scrutinizing the impact residential characteristics have on commute distance.

4 Methodology and Data Sources

For addressing these objectives we follow a purely quantitative approach. We use the third level of the 76th round of NSSO on "Drinking Water, Sanitation and Housing conditions." [The National Sample Survey Organization (NSSO) is a part of the Ministry of Statistics and Programme Implementation (Mospi), Government of India. NSSO has conducted nationwide sample surveys on various socio-economic aspects since 1950. These surveys are conducted in the form of rounds extending normally over a period of one year. The sample is selected through a multi-level random sampling and is considered representative of India's population.

Gender is one of the key socio-demographic variables that can influence travel behaviour, but it is often the least understood. As the focus of the present analysis is to understand the determinants of commuting in women, hence, the households where the women are not participating in commuting are removed from the data. This leaves us with 40745 observations or 38.17% of the original sample.

We begin our analysis by descriptive statistics. This provides a preliminary overview of general behaviour trends on mean travel distance. Subsequent to this analysis, we compare and contrast the commute pattern amongst males and females. This analysis is carried out at two levels namely at the aggregated level and at the disaggregated level, i.e., rural and urban. In addition, we compare the commute distance characteristics of urban and rural females. As the variable is not jointly distributed, we use the cosine similarity for this comparison.

The data set is collated at the household level, i.e., individual characteristics like age, level of education etc., are not collected. Therefore, for the analysis, we use household level socio-economic characteristics. This is in addition to the nature of the residential area and the tenure of the dwelling (Refer Table 1) for characteristics of the variables).

For setting-up the regression equation, we examine the cross-tabulated results between the socio-economic variables and commute trip distance. This analysis will help in understanding whether the socio-economic variables have an impact on commute distance.

Subsequent to this analysis we setup the regression equation. Though the data is ordinal in nature, the categories are exclusive, and hence, we use the stratified regression. Our regression equation takes the form

$$y_i = d_i + b_i x_i + b_i w_i + b_i z_i \tag{1}$$

where y_i is the commute distance category, d_i is the dummy which represents the area type namely urban or rural, x_i represents all the socio-economic variables including the characteristics of the dwelling and b_i is the estimator.

Table 1 Description of variables

Variable		Nature	Level
Dependent		Categorical	8
Predictor	Rural/Urban	Dummy	2
Predictor	Religion	Categorical (with Buddhism as the calibrating level)	8
	Social Group	Categorical (With Other Backward class as the calibrating level)	4
	Tenurial status	Categorical (employee quarter as the base level)	7
	Area of residence	Categorical (with denoted slum as the base level)	4
	Family size	Integer (with minimum value of 1)	
	Usual monthly consumer expenditure	Integer (proxy for income)	

4.1 Is There a Stark Difference Between Male and Female Commute Patterns?

As a primer to our analysis, we begin by examining the commute of men and women across the different distance buckets at an aggregate and disaggregate level, i.e., rural and urban (Refer Table 1). Additionally, we use descriptive statistics for understanding how far men and women commute.

A cursory examination of the commute pattern reveals that more than half the commute trips are for a distance of less than five kilometres and that at an aggregate level '1–5 kms' is the modal and median class for both men and women. At the disaggregated level we observe that for men this characteristic is preserved. Whereas for rural women, the modal class is '1–5 km', while the median class is '<1 km.' In contrast to this for urban women the modal class is 'No distance,' while the median interval is '<1 km.' This means that men and women have different commute patterns. For validating our claims we compute the cosine similarity between these. We propose this measure as the commute trips undertaken by men and women are not jointly distributed (Refer Table 2 for the cosine similarity).

In stark contrast to our earlier conjecture, we observe that the value of cosine similarity carries a positive sign and is greater than 0.7. Therefore, we deduce that there is not much difference between the commute patterns of men and women. The only plausible reason that can be forwarded for these patterns is the social roles that women undertake. Or alternatively, we can deduce that the family setting may be playing a role in determining the commute pattern. On contrasting the commute pattern of rural women against urban women (cosine similarity of 0.9229), we can hypothesize that urban and rural women have similar commute patterns.

4.2 Determinants of Commuting Patterns in Women

In the previous section, it was observed that men and women had statistically similar limits to commute patterns. Socio-economic and demographic characteristics of the households can be the only explanatory variables that can result in such an outcome. Therefore, before setting-up the regression equation, we look at the association of the different categorical variables namely rural/urban, religion, social group, area of residence and tenurial type with the maximum distance the female commutes (Refer Table 3, 4, 5, 6 and 7).

Based on the above table, we inferred that the participation rate of women is higher in rural areas (69.85%). Due to these higher participation rates, rural females are more mobile, i.e., more than six-tenths of the rural women have different work and residence. We observe an inflection point at '10–15 km' beyond which urban women travel more than rural women. Based on the Cramer v we conclude that there is a very weak association between the maximum distance commuted and the binary

Table 2 Distribution of commute trips over the different distance buckets (author's computation)

	No distance	<1 km	1–5 km	5–10 km	10–15 km	15–30 km	>30 km	Modal class	Median class (km)
Aggregate men	8.85	18.62	39.97	14.35	7.79	5.63	4.78	1–5 km	1–5
Aggregate women	34.42	21.26	34.53	5.19	2.26	1.47	0.88	1–5 km	1–5
Rural men	8.94	21.4	43.22	11.8	6.05	4.71	3.87	1–5 km	1–5
Rural women	31.76	23.68	37.55	3.95	1.44	1	0.64	1–5 km	< 1
Urban men	8.67	12.96	33.32	19.56	11.32	7.52	6.64	1–5 km	1–5
Urban women	43.15	13.36	24.63	9.26	4.94	3	1.66	No distance	<1

Table 3 Cosine similarity

	Aggregate men-aggregate women	Rural men-rural women	Urban men-urban women	Rural women-urban women
Cosine similarity	0.8434	0.8836	0.7025	0.9229

Table 4 Commuting patterns in rural and urban women

	No distance	<1 km	1–5 km	5–10 km	10–15 km	15–30 km	>30 km
Rural	23.29	16.88	24.87	2.85	0.97	0.58	0.41
Urban	14.41	4.37	6.75	2.29	1.15	0.69	0.49

Table 5 Maximum distance commuted by women across various religions

	No distance	<1 km	1–5 km	5–10 km	10–15 km	15–30 km	>30 km
Hinduism	27.44	17.07	25.45	3.88	1.63	0.99	0.73
Islam	4.94	1.35	1.59	0.35	0.1	0.08	0.08
Christianity	3.06	1.83	3.27	0.7	0.26	0.12	0.05
Sikhism	0.62	0.07	0.18	0.04	0.04	0.02	0.02
Jainism	0.13	0.03	0.04	0.02	0.01	0.01	0
Buddhism	0.69	0.35	0.48	0.08	0.02	0.01	0.01
Zoroastrianism	0	0	0.01	0	0	0	0
Others	0.82	0.55	0.6	0.08	0.05	0.04	0.01

Table 6 Maximum distance commuted by women across various castes

	Schedule tribe	Schedule caste	Other backward classes	Others
No distance	6.24	5.82	14.58	11.06
<1 km	5.38	3.82	8.35	3.7
1–5 km	7.89	6.18	12.56	4.99
5–10 km	0.98	0.91	2.02	1.23
10–15 km	0.34	0.41	0.8	0.57
15–30 km	0.12	0.28	0.46	0.42
>30 km	0.1	0.16	0.32	0.33

(rural/urban) of 0.2225. We observe that more than a third of the working women reside at the place of their work. Likewise, another thirty percent of the sampled women commute for up to five kilometres.

Table 7 Maximum distance commuted by women across various tenurial status of residence

	No distance	<1 km	1–5 km	5-10 km	10–15 km	15–30 km	>30 km
Owned/ freehold	33.21	19.07	28.52	4.23	1.73	1.06	0.72
lease hold	0.29	0.13	0.32	0.11	0.04	0.02	0.01
hired; employer quarter	0.76	0.33	0.21	0.06	0.04	0.01	0.02
Hired unit with written contract	0.56	0.18	0.43	0.16	0.1	0.06	0.04
Hired unit with no written contract	2.48	1.24	1.85	0.53	0.19	0.12	0.1
No dwelling	0	0.02	0.01	0	0	0	0
Others	0.4	0.29	0.27	0.05	0.03	0	0.01

In terms of religious constitution, Hindu women constitute more than three-fourths (77.19%) of the female workforce. This is followed by women believing in the faith of 'Christianity' (9.29%) and 'Islam' (8.49%). The other religious sects account for less than six percent of the female workforce. More than seven-tents (72.79%) of the women who reside at the place of their work follow the Hindu faith. Likewise, we find that the representative population that follows Zoroastrianism don't commute for more than five kilometres. This result may show a sampling bias. A very weak association between distance commuted and religion of 0.06 was observed.

From the preliminary scrutiny of Table 5, we can deduce that women belonging to the 'Other Backward Class' are ready to commute or conversely, we can presume that the female participation of households belonging to the 'Other Backward Class' is higher. In more than a third of the households, females work at the place of their residence. For understanding whether the maximum distance travelled by the female is impacted by the household's social group, we compute Cramer's V. A weak association (0.0995) is observed.

Though Naess in his seminal work shows that residential characteristics rarely affect commute patterns, we check whether the stated fact is true for Indian women. We compute the association between the maximum distance that the female travelled at the residential tenurial type and the area of the residence (Refer the association between residential tenurial type and maximum distance travelled by the female of the household and area of residence and the maximum distance that the female travels).

We find that more than four-fifths of the women travelling the highest distances for accessing employment live in owned/free-hold households and a third of the total sampled households have females who work and reside in the same place. This shows that fixed tenurial status is highly associated with wage employment. At an aggregate level, we find a very weak association (0.0443) between tenurial status and commute distances. This bolsters the claims of Naess [12]. To validate these claims we also compute the association between area type and limits to commute distance.

From the cursory scrutiny of Table 6, we may infer that the more formal the arrangement, i.e., legal status of an area, the propensity to commute increases. Even then, we observe that more than in third (37.72) of the sampled households the maximum distance that the female commuted was 'No distance' or these women resided at the place of their work. Negligible association of 0.02241 was observed between the area type and the limits to female commute. Based on the above associations, we may construe that other factors like family size, education status, income, etc. may be influencing the limits to commute necessitating a detailed analysis.

The commute distance is an effect of residential self-selection (RSS) [12]. Therefore, we consider each commute distance category as a discrete result arising from the interaction of socio-economic and demographic variables (Refer Table 8 for regression results). For the regression, we use rural, Buddhism, other backward classes, Employer rented and denoted slum as the zero value of the dummy represented by sector, religion, social group, tenure of residence and area type respectively. As income data was not available, we used the monthly expenditure as the proxy for monthly income

In the previous analysis, it was observed that rural and urban commute patterns are similar. Therefore, it could be inferred that the urban/rural dummy would become statistically insignificant. In contrast to this, we observe that the urban/rural dummy is statistically significant across all the seven levels. This difference may be attributed to the weight effect, i.e., the regression results are for the sample.

From the regression results, we infer that 'Religion' shows varying influence. Religion is statistically significant till the '1–5 km' beyond which the variable is insignificant. Further, this significance pertains to certain religions. This meant that the social construct is having an influence on the commute distance. Additionally, these results may suffer from sampling bias, e.g., according to the data 'Parsis' do not travel for more than five kilometres accessing jobs (Refer the cross tabulation of religion-distance (Table 4). Similarly, we observe that both tenure of residence and area of residence have a varying influence on the maximum distance travelled by the female for accessing employment. This result appears to be at variance to the results of the study conducted for Bangalore. Monthly expenditure or income appears to have a strong influence on the maximum distance commuted by the female. In contrast to this family size has a fluctuating influence on the maximum distance that the female commutes, even the signs fluctuate. This meant that education and the nature of employment may be playing an important role in the limits to female commute.

Table 8 .

	No distance	<1 km	1–5 km	5–10 km	10–15 km	15–30 km	>30 km
Notified slum	0.42	0.18	0.37	0.07	0.04	0.01	0.01
Non-notified Slum	0.26	0.16	0.29	0.06	0.03	0.04	0.01
Squatter settlement	0.13	0.1	0.07	0.02	0.01	0	0
Others	36.91	20.8	30.89	4.99	2.03	1.22	0.87

5 Conclusion

Both at an aggregate level and at a disaggregate level, i.e., urban and rural, men and women have statistically similar commute patterns. Additionally, urban and rural women too show very high similarity in their commute patterns. The only possible reasons that we can provide are the social functions that women undertake. This result is important for transport planners, as this shows that the special requirements of women like safety, security etc., can be incorporated in the comprehensive transport policy and a separate transport policy as entailed by feminist scholars is not necessary.

From these results, we may infer that the socio-economic and demographic variables may be playing an important role in determining the limits to commute. In contrast to these assumptions, we find that the household characteristics were weakly associated with the commute distance. These results corroborate the findings of Naess [12].

The regression results provide us an insight into this. The type of region, i.e., urban or rural has a positive impact on the limits to travel. Likewise, income too has an influence on limits to female commute. In contrast to this religion, social group, tenurial status of residence, area of residence and family size have a varying influence. This means that education, skill set that the female possesses and the nature of employment are likely to have an influence on the limits to female commute.

The present data set was collected at the household level therefore, details about the mode of commute, level of education that the female has attained etc. were not available. This may have had an influence on the model. As an extension to this study, we will investigate whether the limits to commute and income follow an inverted Kuznets curve. Additionally, we intend to separate wage employment from other kinds of employment and check the constancy of the relation.

Appendix

See Table 9.

Table 9 Results of regression

Dependent variable	Intercept	Sector_Urban	Religion_christian	Religion_hindu	Religion_islam	Religion_jain	Religion_other
No travel	−4.163e-01 (1.577e-01)**	7.258e-01 (2.749e-02)***	−4.222e-01 (8.802e-02)***	−5.536e-01 (8.180e-02)***	1.913e-01 (8.958e-02)*	−1.600e-01 (2.244e-01)	−3.520e-01 (1.072e-01)**
<1 km	−7.818e-01 (1.951e-01)***	−5.743e-01 (3.610e-02)***	−8.580e-02 (1.051e-01)	2.078e-01 (9.783e-02)*	−8.453e-02 (1.093e-01)	−1.935e-01 (3.392e-01)	2.911e-01 (1.240e-01)*
1–5 km	−8.698e-01 (1.825e-01)***	−5.908e-01 (3.104e-02)***	3.304e-01 (9.367e-02)***	3.233e-01 (8.805e-02)***	−2.859e-01 (9.909e-02)**	−8.655e-02 (2.861e-01)	5.141e-02 (1.156e-01)
5–10 km	−9.253e-06 (2.220e-06)***	4.139e-01 (5.529e-02)***	4.800e-01 (1.974e-01)*	−5.158e-02 (1.893e-01)	−3.087e-01 (2.089e-01)	−1.111e-01 (4.426e-01)	−2.722e-01 (2.595e-01)
10–15 km	−4.590e+00 (5.169e-01)***	7.076e-01 (8.255e-02)***	8.627e-01 (3.738e-01)*	4.461e-01 (3.625e-01)	−1.739e-01 (3.971e-01)	4.309e-01 (6.371e-01)	7.298e-01 (4.220e-01)
15–30 km	−6.205e+00 (7.745e-01)***	5.924e-01 (1.055e-01)***	7.590e-01 (4.807e-01)	2.358e-01 (4.574e-01)	−1.429e-01 (4.932e-01)	8.755e-01 (6.637e-01)	9.301e-01 (5.249e-01)
>30 km	−5.874e+00 (8.066e-01)***	7.346e-01 (1.246e-01)***	2.054e-01 (6.228e-01)	4.847e-01 (5.876e-01)	3.147e-01 (6.158e-01)	4.802e-01 (9.330e-01)	−3.745e-01 (8.211e-01)

Dependent variable	Religion_parsi	Religion_sikh	Group_others	Group_sc	Group_st	Tenurial_lease	Tenurial_ non-written
No travel	−1.012e+00 (8.537e-01)	5.754e-01 (1.322e-01)***	4.113e-01 (2.789e-02)***	−1.311e-01 (3.101e-02)	−2.864e-01 (3.394e-02)***	−6.372e-01 (1.429e-01)***	−6.705e-01 (9.478e-02)***
<1 km	−3.053e-01 (1.095e+00)	−9.924e-01 (2.195e-01)***	−1.639e-01 (3.525e-02)***	−1.209e-02 (3.535e-02)	2.498e-01 (3.614e-02)***	−8.746e-01 (1.818e-01)***	−1.687e-01 (1.123e-01)
1–5 km	8.327e-01 (7.811e-01)	−3.336e-01 (1.563e-01)*	−3.499e-01 (3.139e-02)***	6.899e-02 (3.079e-02)*	1.586e-01 (3.268e-02)***	8.607e-01 (1.626e-01)***	8.793e-01 (1.268e-01)***

(continued)

Table 9 (continued)

Dependent variable	Religion_parsi	Religion_sikh	Group_others	Group_sc	Group_st	Tenurial_lease	Tenurial_ non-written
5–10 km	1.086e+00 (1.108e+00)	−5.092e-01 (3.205e-01)	−1.172e-01 (6.074e-02)	1.784e-02 (6.571e-02)	−3.277e-01 (7.730e-02)***	1.230e+00 (2.587e-01)***	6.065e-01 (2.142e-01)**
10–15 km	−9.190e+00 (1.979e +02)	8.926e-01 (4.393e-01)*	−1.152e-01 (9.222e-02)	1.451e-01 (9.856e-02)	−3.793e-01 (1.245e-01)**	8.379e-01 (3.651e-01)*	1.359e-01 (2.884e-01)
10–15 km	−8.747e+00 (1.948e +02)	3.973e-01 (5.665e-01)	1.100e-01 (1.135e-01)	3.246e-01 (1.230e-01)**	−1.022e+00 (1.932e-01)***	1.171e+00 (5.363e-01)*	6.247e-01 (4.384e-01)
>30 km	−1.149e+01 (8.922e+02)	7.794e-01 (7.031e-01)	3.605e-01 (1.291e-01)**	1.435e-01 (1.568e-01)	−3.636e-01 (2.107e-01)	−3.770e-01 (6.731e-01)	4.858e-02 (3.732e-01)

Dependent variable	Tenurial_others	Tenurial_owned	Tenurial_written	Area_ not-denoted	Area_others	Area_squatter	Size	Monthly
No travel	−4.985e-01 (1.337e-01)***	−3.148e-01 (8.684e-02)***	−7.914e-01 (1.205e-01)***	−3.006e-01 (1.560e-01)	5.286e-01 (1.016e-01)***	8.867e-02 (2.069e-01)	3.033e-02 (5.848e-03)***	−1.838e-05 (1.937e-06)***
<1 km	1.244e-01 (1.491e-01)	−4.112e-01 (1.025e-01)***	−5.991e-01 (1.610e-01)***	6.231e-02 (1.904e-01)	−1.412e-01 (1.320e-01)	7.989e-01 (2.280e-01)***	2.307e-02 (6.937e-03)***	−1.318e-05 (2.652e-06)***
1–5 km	5.002e-01 (1.637e-01)**	6.824e-01 (1.201e-01)***	1.055e +00 (1.496e-01)***	−2.035e-03 (1.532e-01)	−5.981e-01 (1.040e-01)***	−6.122e-01 (2.349e-01)**	1.192e-03 (6.162e-03)	−9.253e-06 (2.220e-06)***
5–10 km	2.333e-01 (3.033e-01)	3.449e-01 (2.046e-01)	7.221e-01 (2.416e-01)**	1.568e-01 (2.888e-01)	1.385e-01 (1.944e-01)	−1.361e-01 (4.342e-01)	−1.859e-02 (1.196e-02)	2.608e-05 (2.832e-06)***
10–15 km	1.988e-01 (4.069e-01)	2.289e-01 (2.684e-01)	6.718e-01 (3.121e-01)*	5.199e-02 (3.736e-01)	−1.678e-01 (2.488e-01)	−2.409e-01 (5.655e-01)	−9.764e-02 (1.929e-02)***	3.732e-05 (3.472e-06)***
10–15 km	−6.409e-01 (8.247e-01)	6.708e-01 (4.174e-01)	9.954e-01 (4.646e-01)*	1.643e+00 (5.240e-01)**	6.118e-01 (4.572e-01)	−2.945e-01 (1.103e+00)	−9.131e-02 (2.389e-02)***	4.206e-05 (3.865e-06)***
>30 km	−2.299e-01 (6.100e-01)	−2.172e-01 (3.462e-01)	2.562e-01 (4.238e-01)	3.177e-02 (6.514e-01)	1.511e-01 (4.205e-01)	−1.220e+01 (2.031e+02)	2.751e-02 (2.462e-02)	2.253e-05 (4.952e-06)***

***99.99% Confidence interval
**99% Confidence interval
*95% Confidence interval

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